Do **not** write solutions on this page.

11. [Maximum mark: 17]

The lengths, in metres, of jumps in a long jump competition can be modelled by a continuous random variable X with probability density function f defined by:

$$f(x)=\begin{cases} 0, & x<3 \\ a(x-3)^3+b(x-3)^2, & 3\le x\le 9 \\ 0, & x>9 \end{cases}$$

where a, $b \in \mathbb{R}$, a, $b \neq 0$.

(a) Show that $324a+72b=1$. [4]

It is given that $f(9)=0$.

(b) (i) Show that $6a+b=0$.

(ii) Determine the value of a and the value of b. [4]

(c) Show that the median of X is less than the mode of X. [5]

Matt has the final jump in the competition. He needs to jump at least $8.52\,\mathrm{m}$ to win the competition.

(d) Given that he jumps over $8\,\mathrm{m}$, use the model to find the probability that he wins the competition. [4]